

STOCKHOLM VASTERAS, Sweden

WMO: 024460

Lat: 59.589N

Lon: 16.634E

Elev: 21

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.6	5.0	-5.7	4.1	-0.3	0.0	5.5	6.4	25.3	36.8	22.8	36.3	2.1	0	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	81.0	64.8	78.4	63.5	75.1	62.1	67.3	77.3	65.4	74.7	63.7	72.1	8.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.1	89.5	70.6	62.2	83.8	69.2	60.4	78.6	67.9	31.7	77.0	30.3	74.7	29.0	71.9	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.8	18.5	16.5	DB	-6.7	85.4	8.4	3.2	-12.8	87.7	-17.7	89.6	-22.4	91.4	-28.5	93.8	(4)
			WB	-7.2	69.6	8.3	2.4	-13.2	71.4	-18.1	72.8	-22.7	74.2	-28.7	75.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.0	26.2	26.4	32.6	42.0	50.8	58.7	63.6	61.7	53.5	44.6	36.0	30.2
	DBStd	14.98	9.75	9.70	7.68	6.14	7.02	5.27	4.94	4.98	5.38	6.17	7.26	10.03
	HDD50	3517	739	661	538	250	76	3	0	1	28	189	419	613
	HDD65	7754	1204	1081	1003	689	441	198	88	124	346	633	869	1078
	CDD50	1314	0	0	0	10	101	265	420	364	133	20	1	0
	CDD65	78	0	0	0	0	1	10	43	23	1	0	0	0
	CDH74	867	0	0	0	0	40	138	473	214	2	0	0	0
	CDH80	147	0	0	0	0	1	13	104	29	0	0	0	0
Wind	WSAvg	7.8	8.2	7.9	8.1	7.9	7.9	7.6	7.1	7.0	7.6	8.1	8.3	8.2
Precipitation	PrecAvg	21.4	1.3	1.0	1.1	1.2	1.4	1.9	2.9	2.8	2.3	2.0	2.0	1.5
	PrecMax	31.4	2.5	2.2	2.2	2.8	3.7	4.4	5.9	6.1	7.6	4.3	3.8	4.6
	PrecMin	15.7	0.2	0.2	0.0	0.0	0.4	0.2	0.9	0.3	0.6	0.4	0.7	0.4
	PrecStd	3.7	0.6	0.6	0.6	0.7	0.9	1.0	1.4	1.5	1.5	1.2	0.9	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.8	46.6	56.9	68.4	78.5	82.1	87.6	83.6	73.4	61.1	53.6	50.3
		MCWB	42.7	41.3	46.9	52.5	61.5	63.5	67.8	65.2	60.9	56.7	50.7	47.1
	2%	DB	42.6	43.0	50.3	64.0	73.8	78.4	82.7	79.1	69.6	58.7	50.4	46.8
		MCWB	40.5	39.5	42.7	50.1	58.3	62.3	66.2	64.5	60.3	54.5	48.5	44.3
	5%	DB	40.6	40.8	46.5	58.8	69.7	73.8	79.2	75.9	66.0	55.8	48.3	44.5
		MCWB	38.8	38.0	40.6	48.2	56.5	60.2	64.4	63.2	58.1	52.6	46.9	42.1
	10%	DB	37.7	38.8	42.9	54.0	64.6	70.1	75.5	73.0	62.9	53.9	46.4	41.4
		MCWB	36.0	36.5	38.5	44.9	53.3	58.6	63.3	61.7	56.6	50.8	44.9	39.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.8	42.6	46.8	54.7	63.8	66.6	70.3	69.0	63.1	57.9	51.4	48.0
		MCDB	43.7	44.5	53.8	66.0	74.6	76.4	80.2	78.0	69.0	60.5	53.1	50.6
	2%	WB	40.3	39.9	43.5	51.1	60.3	63.9	68.3	66.6	60.9	55.6	49.1	44.6
		MCDB	41.4	42.0	48.8	60.6	69.6	74.3	79.1	75.0	67.0	57.7	50.3	46.2
	5%	WB	38.5	37.8	41.3	48.9	57.4	61.8	66.4	64.7	59.2	53.8	46.6	41.8
		MCDB	39.8	39.6	45.0	57.2	67.7	71.4	76.7	72.5	64.5	56.1	47.9	43.3
	10%	WB	36.2	36.2	39.0	46.3	54.8	59.9	64.5	63.2	57.4	51.8	44.7	39.7
		MCDB	37.5	37.9	42.2	52.6	63.6	68.7	73.8	70.3	62.1	54.0	45.6	41.1
Mean Daily Temperature Range	5% DB	MDBR	9.5	11.5	13.6	18.6	19.5	18.2	18.8	18.4	16.1	11.7	8.3	8.9
		MCDBR	8.3	10.6	17.9	26.7	27.9	25.7	26.1	24.1	20.5	12.8	9.6	9.4
		MCWBR	8.0	8.2	12.0	15.6	15.2	13.2	12.2	11.7	12.4	9.9	8.8	8.7
	5% WB	MCDBR	8.1	9.4	15.9	23.9	24.4	21.8	23.0	20.2	18.0	11.3	9.4	8.7
		MCWBR	8.1	8.0	11.4	14.8	14.2	12.2	12.0	10.7	11.5	9.4	8.8	8.3
Clear-Sky Solar Irradiance	taub	0.274	0.316	0.338	0.374	0.371	0.361	0.386	0.380	0.344	0.332	0.301	0.257	
	taud	2.203	2.270	2.352	2.319	2.392	2.441	2.392	2.406	2.492	2.424	2.263	2.131	
	Ebn at Noon	164	218	252	262	272	277	266	258	250	213	152	122	
	Edh at Noon	14	22	27	34	34	33	34	31	24	19	13	10	
All-Sky Solar Radiation	RadAvg	98	302	781	1246	1611	1802	1652	1307	857	384	124	64	
	RadStd	16	43	99	122	149	114	164	131	98	63	28	6	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-218	-265	N/A	N/A

Nomenclature: See separate page